Research Statement

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My research interests are located at the interplay of enumerative combinatorics and constructive algebra and cover algorithmic enumerative combinatorics, effective algebraic geometry, differential algebra and dynamical systems of algebraic correspondences. Some relevant keywords in the context of enumerative combinatorics are *lattice walks*, *generating functions*, *discrete differential equations* and *kernel method*. In the following I give an overview of some of my research highlights. For a complete list of papers and preprints I refer to my CV.

Discrete differential equations are functional equations that arise in the enumeration of e.g. maps and restricted lattice walks. The typical question is whether their solutions are algebraic, D-finite or D-algebraic and how they can be expressed in terms of algebraic or differential equations. These expressions can then be used to answer various questions about the solutions' coefficients using methods from computer algebra and analytic combinatorics.

Discrete differential equations that arise in the enumeration of restricted lattice walks have recently received much attention and there are meanwhile various methods they can be studied with. The methods may be algebraic and / or analytic. They may rely on elementary power series algebra and computer algebra, complex analysis and boundary value problems, Galois theory of difference equations, analytic combinatorics or probability theory. Some of these methods are well-understood. However, many of those that rely essentially only on elementary power series algebra are not. They were often developed to solve specific functional equations. Thus many of the arguments they rely on were introduced only adhoc. The question how these methods generalize is subject of much of my research.

1 Kernel method(s) – a survey [8]

A survey that covers many of these elementary methods is given in [8]. It includes, but is not limited to, the classical kernel method, the orbit-sum method, and the invariant method, which have already received considerable attention in my work. To make things more concrete, let me illustrate these methods, and thereby motivate some of the research questions we could address.

2 Classical kernel method [11]

Example 1. The functional equation for the generating function of simple walks on the half-line is

$$x(1-t(x^{-1}+x))F(x;t)=x-tF(0;t).$$

Following [14, Exercise 2.2.1.-4], it can be solved by eliminating $F(x;t)$ by coupling x and t without altering $F(0;t)$. Indeed, viewed as a polynomial in x, the kernel polynomial

$$K(x,t):=x(1-t(x^{-1}+x))$$

has two roots,

$$x_0(t)=\frac{1-\sqrt{1-4t^2}}{2t} \quad \text{and} \quad x_1(t)=\frac{1+\sqrt{1-4t^2}}{2t},$$

only one of which, namely $x_0(t)$, can be interpreted as an element of $\mathbb{Q}[[t]]$. So the composition $F(x_0(t);t)$ is well-defined and the substitution of $x_0(t)$ for x in (1) results in

$$0=x_0(t)-tF(0;t).$$

So

$$F(0;t)=\frac{x_0(t)}{t} \quad \text{and} \quad F(x;t)=\frac{1-x^{-1}x_0(t)}{1-t(x^{-1}+x)}.$$

The method just employed is called the *classical kernel method.* It was originally introduced by Donald Knuth [14] and later generalized by Bousquet-Mélou and Jehanne in [4], showing that solutions of ordinary (not necessarily linear) discrete differential equations are algebraic. Motivated by the enumeration of inhomogeneous lattice walks, we generalized the method to the setting of systems of linear discrete differential equations and showed that their solutions are algebraic too [11]. The results were extended to systems of non-linear discrete differential equations in [15].

3 Orbit-sum method [12]

Example 2. Equation (1) can also be solved by exploiting that $1-t(x^{-1}+x)$ is invariant under $x \mapsto x^{-1}$. Applying the transformation to equation (1) gives

$$x^{-1}(1-t(x^{-1}+x))F(x^{-1};t) = x^{-1} - tF(0;t),$$

and subtracting the latter from the former and dividing by $x(1-t(x^{-1}+x))$ results in

$$F(x;t) - x^{-2}F(x^{-1};t) = \frac{1-x^{-2}}{1-t(x^{-1}+x)}.$$

As $F(x;t)$ involves only non-negative powers of x, while $x^{-2}F(x^{-1};t)$ involves only negative ones, we can eliminate $x^{-2}F(x^{-1};t)$ by applying $[x^{\geq}]$ and conclude that

$$F(x;t) = [x^{\geq}]\frac{1-x^{-2}}{1-t(x^{-1}+x)}.$$

The non-negative part of a rational series is D-finite. Hence so is $F(x;t)$.

The *orbit-sum method* just illustrated and originally invented in [5] proceeds in three steps: In the first step, a set of substitutions that can be applied to the given functional equation, the so-called orbit, is determined. In the second step, a linear combination of the various transformed versions of the equation is formed with the goal of eliminating all the evaluations of F. The resulting equation(s) then contain only the unknown series F and various series obtained from the substitutions. In the third step of the method, by means of applying the operator $[x^{\geq}]$, which discards terms with negative powers in x, an expression for the unknown series F is obtained.

Due to the simplicity of the substitutions, each of these steps appears to be trivial in the above example. But, in general, none of them is, as the substitutions do not need to be rational but only algebraic. Moreover, they are only given by their minimal polynomials [3]. In [12] we address the difficulties caused by that. We explain how to construct an algebraic function field that contains all the algebraic functions appearing in the orbit, and how to do computations therein. This can be done on the level of "formal" algebraic extensions. The third step, however, crucially depends on series interpretations of the algebraic functions and an estimate of their support. We explain how to embed the algebraic function field into suitably chosen fields of series. The question is then whether for the equation at hand there exists an embedding that allows the orbit-sum method to conclude. To answer this question, we offer an algorithmic sufficient condition based on the computation of the supports of the series involved.

4 Newton-Puiseux algorithm [9]

Much of the reasoning in [12] relies on the Newton-Puiseux algorithm, an algorithm for computing series solutions of polynomial equations. The algorithm is classical and well-known when dealing with equations in two variables. The multivariate version, however, has been invented only in the 90's of the last century, and there are important questions that have been discussed only recently. In [9] I address some of them. For instance, I explain how the Newton-Puiseux algorithm can be used to encode algebraic series and how to do effective arithmetic on the level of these encodings. I furthermore discuss how to derive information about the convex hull of the support of algebraic series.

5 Invariant method – Invariants and decoupling functions [13, 7, 6]

Example 3. The symmetry of the kernel polynomial that the orbit-sum method is based on is not the only symmetry that can be exploited to solve the kernel equation. There is also a symmetry that appears in $K(x,t)$ having a (non-trivial) rational multiple in $\mathbb{Q}(x)+\mathbb{Q}(t)$. One such multiple is

$$-\frac{K(x,t)}{xt} = x^{-1} + x - \frac{1}{t}. \tag{1}$$

Following [1], it can be used to eliminate all terms involving x from the right-hand side of equation (1) by subtracting equation (1) and adding

$$K(x,t)\frac{F(x;t)}{xtF(0;t)} = \frac{1}{tF(0;t)} - x^{-1}.$$

Multiplication of the resulting equation by $tF(0;t)$ gives

$$K(x,t)\left(-x^{-1}F(0;t) - tF(0;t)F(x;t) + x^{-1}F(x;t)\right) = 1 - F(0;t) + t^2F(0;t)^2.$$

The valuations of $K(x,t)$ are zero with respect to both x and t, and so are the valuations of $1-F(0;t)+t^2F(0;t)^2$, if we assume the expression to be non-zero. Hence the valuations of the coefficient of $K(x,t)$ on the left-hand side of the equation are zero too, and the right-hand side is a multiple of $K(X,t)$ in $\mathbb{Q}[[x,t]]$ that does not depend on x. A contradiction. So the assumption is wrong and

$$1 - F(0;t) + t^2 F(0;t)^2 = 0.$$

So $F(0;t)$ is algebraic, and so is $F(x;t)$.

The *invariant method* just illustrated relies on the polynomial $K(x,t)$ having a non-trivial rational multiple in $\mathbb{Q}(x) + \mathbb{Q}(t)$. Whether for a given polynomial such a multiple exists, and in case it does, how to construct it, are some of the questions it raises. Based on [13], we answered the latter in [7]. The main result is a (semi-)algorithm that takes a bivariate polynomial and returns the generator of a field $\mathrm{F}(K)$ that describes all its separated multiples.

Another question raised by the invariant method is whether a given bivariate rational function $r(x,t)$ is the sum of two univariate rational functions $f(x)$, $g(t)$, modulo a given polynomial $K(x,t)$. In [6] we provide an algorithm for the case that K has a non-trivial separated multiple and a (conjectural) semi-algorithm when this is not the case.

6 Algebraic correspondences [10]

As indicated before, some of the algorithms are just semi-algorithms: they do not terminate on any input. Their termination depends on properties of an algebraic correspondence associated with $K(x,t)$ and raises several problems and questions. Here are three of them:

- Given a point, decide whether its orbit is infinite.
- Are there only finitely many orbits, when there is an infinite orbit?
- Given two points, decide whether one of them is contained in the orbit of the other.

The second question / problem is discussed in [10]. The others, however, are still open.

7 Invariant method – a differential analogue [ongoing work]

The applicability of the invariant method depends on the existence of certain rational functions which are called invariants and decouplings functions. It is known that for certain discrete differential equations their solutions are D-finite if and only if there are invariants and that they are algebraic if and only if there are not only invariants but also decoupling functions. The invariant method as presented in [1, 2] provides an elementary way to construct an annihilating polynomial for their solutions from these invariants and decoupling functions. In ongoing work with Charlotte Hardouin it is explained how the invariant method can be modified to construct differential equations for their solutions when there are only invariants. The following example gives a hint of how this can be done.

Example 4. Let us again consider equation (1). It is easy to derive a linear differential equation for $F(x;t)$ in x. We just need to differentiate both sides of the equation with respect to it. The result is a differential equation for $F(X;t)$ that is inhomogeneous. By differentiating another time we can get rid of the inhomogeneity and find that

$$(-t+x-tx^2)\frac{\partial^2 F(x;t)}{\partial x^2}+2(1-2tx)\frac{\partial F(x;t)}{\partial x}-2tF(x;t)=0.$$

The construction of a differential equation for $F(x;t)$ with respect to t is slightly more involved. We claim that the invariant associated with $K(x,t)$ gives rise to a differential operator on the function field of the curve associated with $K(x,t)$ that annihilates the image of x and maps the image of t to something that does not involve the image of x. It is not too difficult to see that the operator

$$L=\left(\frac{1}{t^2}-4\right)da^2-\frac{1}{t}da-1$$

in the derivation

$$da=\frac{x^2}{1-x^2}\frac{\partial}{\partial x}+t^2\frac{\partial}{\partial t}$$

induces such an operator. Applying L to both sides of equation (1) and reordering terms shows that

$$t^3(1-4t^2)\frac{\partial^2 F(0;t)}{\partial t^2}+t^2(3-16t^2)\frac{\partial F(0;t)}{\partial t}-8t^3F(0;t)=0$$

holds up to an additive term that is a multiple of K in $\mathbb{Q}[[x,t]]$. The invariant lemma implies that this additive term is actually zero. Hence $F(0;t)$ is D-finite, and L is an annihilating differential operator. A differential equation in t for $F(x;t)$ can be found by using closure properties of D-finite functions.

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